

Lost in Transcription: Investigating the Role of Audio Information for Video-based Visual Statement Verification

Michela Deodati^{1,*†}, Mohd Kaif^{2,†}, Davide Testa^{3,4}, Giovanni Bonetta³, Alessandro Lenci¹, Oswald Lanz², Raffaella Bernardi² and Bernardo Magnini³

¹University of Pisa, Italy

²Free University of Bozen-Bolzano, Italy

³Fondazione Bruno Kessler (FBK), Trento, Italy

⁴University of Rome La Sapienza, Rome, Italy

Abstract

As new generative AI models are capable of managing rich multimodal content (including text, images, video and audio), there is a growing interest in understanding how such modalities interact and contribute in specific scenarios. In the paper, we present a preliminary investigation into the role of audio information in a video-based Visual Statement Verification task, where a model is given a question about a video and must select the correct statement from a minimally contrastive true–false pair. Given a corpus of videos representing typical everyday activities (e.g., a dinner in a pizzeria, a concert, etc.), we examine how audio information (speech, background sounds, or noise) affects the capacity of a model to reason about the content of the video. We investigate audio through two complementary settings: (i) raw audio directly processed by an omnimodal model; and (ii) audio-derived transcriptions provided as auxiliary textual input to a vision-language model. We conduct controlled experiments and ablation studies on the visually grounded MAIA dataset, categorizing videos according to their acoustic content. Results confirm the expected dominance of visual information over audio, even for videos containing intelligible dialogue, and show that the contribution of audio depends both on visual availability and on whether it is provided as a raw signal or as text. Raw audio provides a substantial benefit only when visual evidence is absent, while its contribution becomes small and non-robust when even minimal visual information is available. These findings highlight persistent challenges in balancing modalities and suppressing auxiliary signals that are not informative for the task.

Keywords

Multimodality, Multimodal Evaluation, Audio, Transcriptions, Benchmarking, Reasoning, VLMs

1. Introduction

In recent years, multimodal AI research has been largely driven by the development of Vision-Language Models (VLMs), which extend the capabilities of language models by integrating textual prompts with visual information from images and videos. Visual features can be effectively connected to language-model representations, enabling a broad range of tasks including image captioning, visual question answering, and instruction following [1, 2]. However, although these models are commonly described as multimodal, their interaction with the external world has traditionally been centered primarily on the integration of vision and language.

Recent advances are extending multimodal modeling beyond the traditional vision-language setting by incorporating audio as an additional input channel, creating what has started being called as Omnimodal Large Language Models. As a result, models can increasingly process richer real-world scenarios in which textual, visual, and acoustic signals jointly contribute to the interpretation of dynamic content

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*Corresponding author.

†These authors contributed equally.

✉ m.deodati@studenti.unipi.it (M. Deodati); mohd.kaif@student.unibz.it (M. Kaif); dtesta@fbk.eu (D. Testa); gbonetta@fbk.eu (G. Bonetta); alessandro.lenci@unipi.it (A. Lenci); oswald.lanz@unibz.it (O. Lanz); raffaella.bernardi@unibz.it (R. Bernardi); magnini@fbk.eu (B. Magnini)

ORCID 0009-0002-6464-8749 (M. Kaif); 0009-0002-2489-5323 (D. Testa); 0000-0003-4498-1026 (G. Bonetta); 0000-0001-5790-4308 (A. Lenci); 0000-0003-4793-4276 (O. Lanz); 0000-0002-3423-1208 (R. Bernardi); 0000-0002-0740-5778 (B. Magnini)



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[3, 4, 5]. At the same time, new benchmarks are beginning to evaluate video understanding under richer modality configurations, including both raw audio and subtitles [6, 7]. These developments make it increasingly important to investigate not only whether models can accept multiple modalities as input, but also how these modalities interact during reasoning.

The question is especially relevant for video understanding. Compared with static images, videos require models to preserve and integrate information across sequences of events. The addition of audio may provide complementary temporal and semantic cues, such as dialogue, background sounds, or acoustic signals associated with ongoing events. However, it also introduces an additional layer of complexity: the model must identify the relevant information in each modality and combine it without being distracted by redundant or weakly related signals. Such an integration cannot be taken for granted, as *omnimodal* models may still exhibit a substantial gap between their vision-language and vision-audio capabilities [8].

In the paper, we present a preliminary investigation into the contribution of audio information to video-based multimodal reasoning. To this end, we leverage the data available in the MAIA benchmark [9, 10] and focus on its Visual Statement Verification (VSV) task, in which a model is presented with a video and must identify the correct statement from a minimally contrastive true–false pair. Such benchmark was originally designed around the visual content of its videos: both the benchmark construction and the VSV task rely on visually grounded information, while the audio track was not considered when creating the evaluation items. Nevertheless, as the audio track remains part of the overall informational content of the videos, we use this setting as a controlled testbed for an initial exploration of how acoustic signals interact with rich visual evidence in current multimodal architectures. Rather than offering a comprehensive evaluation of audio-visual reasoning, our study uses this existing setting to obtain preliminary evidence on the extent to which audio is exploited, ignored, or potentially acts as an interfering signal when combined with visual and textual information. Accordingly, we address the following research questions in a highly controlled multimodal environment:

- **RQ1:** To what extent does raw audio affect video-based VSV when it is jointly processed with visual and textual information?
- **RQ2:** How effectively do multimodal models integrate visual, textual, and auditory information when performing video-based reasoning and understanding tasks?

To address these research questions, we make three main **contributions**. First, we extend the original MAIA evaluation setting by enriching the videos with an auxiliary audio-transcription layer, including both transcribed content and a coarse characterization of the acoustic context of each video. Second, we examine audio from two complementary perspectives: as an additional textual signal obtained through automatic transcription and as a raw acoustic modality directly processed by an audio-vision language model. Third, we evaluate the impact of audio through a set of controlled experimental configurations and targeted ablation studies, investigating how visual, textual, and auditory signals interact under different levels of visual availability.

2. Related Work

The evaluation of multimodal models has traditionally relied on behavioral analyses, in which model capabilities are inferred from their predictions on downstream tasks, including visual question answering and image- or video-based reasoning [11, 12, 13, 14]. Alongside task-oriented evaluations, diagnostic benchmarks have been developed to assess whether models can ground linguistic information in visual evidence rather than merely exploit textual regularities [15, 16, 17]. In particular, controlled evaluations based on minimally contrastive caption–foil pairs make it possible to target specific linguistic and visual phenomena, providing a more fine-grained perspective than aggregate task-level scores [15, 17, 10].

A central issue in this line of research concerns the actual contribution of each modality to the final prediction. The availability of multiple input channels does not necessarily imply that a model uses them in a balanced manner. Multimodal models may rely on distributional biases or unimodal shortcuts,

exploiting information from one modality while largely disregarding the others [18]. Previous work has therefore argued that multimodality should be defined relative to the information that is relevant for a specific task and has proposed methods to quantify the contribution of individual modalities to model predictions [19, 20]. This line of research has highlighted the phenomenon of *unimodal collapse*, which occurs when a unimodal system achieves performance comparable to that of its multimodal counterpart, suggesting that the additional modality is not effectively contributing to the decision process.

The recent integration of raw audio into the VLM computation introduces an additional level of complexity. While traditional video-language models primarily combine visual content and text, emerging approaches increasingly incorporate acoustic information as a distinct input modality. Different strategies have been explored, including the alignment of vision, audio, and language representations in a shared space, the joint processing of visual, auditory, subtitle, and textual information, and the mapping of acoustic signals into soft prompts for language or VLMs [21, 22, 23]. These developments enable models to process richer real-world scenarios in which relevant information may be distributed across visual events, spoken language, background sounds, and other acoustic cues.

In parallel, recent benchmarks have started to evaluate video understanding under richer modality configurations, explicitly incorporating audio and subtitles or targeting temporally localized events involving visual, acoustic, and linguistic information [6, 7]. However, the inclusion of additional modalities does not guarantee that they are integrated effectively. Existing evidence indicates that current *omnimodal models* may exhibit a gap between their vision-language and vision-audio capabilities, suggesting that acoustic information is not always exploited as effectively as textual input [8].

While previous work demonstrates the potential value of audio for video understanding, its specific contribution remains comparatively underexplored in controlled reasoning settings. In particular, it is still unclear whether audio acts as a complementary semantic signal, as a partial substitute for visual evidence, or as a source of interference when multiple modalities are jointly available. Our work addresses this question by systematically varying the information provided to the model and examining the role of both raw audio and audio-derived transcriptions.

3. Experimental Data

For the experiments in this study, we adopt MAIA [9, 10], a native-Italian benchmark designed to assess multimodal reasoning in video-based scenarios. MAIA was developed as a competence-oriented resource in which video understanding is evaluated across several fine-grained reasoning categories, specifically designed to probe whether models can ground language in visual evidence. These categories target whether VLMs can ground linguistic information in dynamic visual evidence and reason over events, entities, actions, spatial configurations, temporal relations, implicit information, and uncertainty. Unlike benchmarks centered on static images or generic task performance, MAIA focuses on realistic video understanding, with content reflecting culturally situated Italian scenarios – places, events, and activities such as a dinner in a pizzeria or a concert – rather than static conversational scenes, making the visual channel the primary carrier of task-relevant information.

The benchmark is built from a curated collection of Italian videos, each associated with 2,400 questions targeting specific reasoning phenomena. Each question is paired with a pool of eight human-written answers, which are then used to generate true declarative statements (captions), visually grounded in the corresponding video. For each caption, a minimally contrastive false statement (foil) is produced by modifying the information relevant to the targeted reasoning category while preserving most of the original linguistic structure. For example, in a spatial case, the caption *At the end of the scene, the little girl walks on a carpet* is contrasted with the foil *At the end of the scene, the little girl walks on a path*. Here, the minimal contrast concerns the spatial information expressed by the final phrase: changing the location from *on a carpet* to *on a path* preserves the overall sentence structure while making the foil visually unsupported by the video. This caption-foil construction makes MAIA suitable for controlled evaluations of visually grounded understanding, as the contrast between alternatives depends on fine-grained visual and semantic evidence.

3.1. Data Enrichment with Audio Transcriptions

We constructed an auxiliary textual representation to enrich each video instance with a structured textual representation of its audio track by using two automatic systems, *Whisper-1* [24] and *GPT-4o-transcribe* [25], in a two-step experimental pipeline (see Appendix A for more detailed information).¹

Whisper-1. The model was used as a standard automatic speech recognition model, without task-specific prompting. We requested that the model return a comprehensive transcription alongside detailed segment-level diagnostic data to produce global indicators that reflect the overall confidence in the transcription. Each output, according to the indicators, was categorized into one of the following contextual classes:

- **MUSIC.** Instances characterized by repetitive vocalizations, onomatopoeic patterns, strongly repetitive tokens, or explicit lexical cues associated with non-speech audio.
- **SILENCE/NOISE.** Videos with empty or highly unreliable transcriptions, especially when accompanied by low log-probability values and relatively high no-speech probability.
- **DIALOGUE.** Videos whose transcriptions displayed sufficiently coherent linguistic content, acceptable log-probability values, and limited evidence of no-speech segments.
- **UNKNOWN_DIALOGUE.** Residual cases that still appeared speech-related but did not satisfy the dialogue criteria with sufficient confidence were assigned to this category, thus preserving potentially informative content while explicitly marking it as uncertain.

GPT-4o-transcribe. It was prompted to assign a contextual class and produce a transcription only when the speech was sufficiently clear, together with a global confidence score.² The predictions were then mapped to the same four contextual classes used above.

Merging Whisper-1 and GPT-4o-transcribe outputs. The outputs of the two models were merged through a dedicated decision procedure. When both systems assigned the same contextual class, it was retained as the final label. For *Dialogue* and *Unknown_Dialogue* cases, transcriptions were semantically reconciled: near-duplicates were consolidated by retaining the more informative version, while complementary sentences were merged after removing redundant content. When the models disagreed, the final label was selected from the system with the higher reliability score.

The resulting distribution of contextual labels is reported in Table 1.

Classification	Number of videos	Percentage
Dialogue	35	35%
Unknown_dialogue	12	12%
Music	17	17%
Silence/Noise	36	36%

Table 1

Distribution of the final audio-transcription classes assigned to the 100 videos after manual revision.

4. Experimental Design

This section presents the experiments designed to assess the contribution of audio information to video-based VSV, examining it both as a raw acoustic signal and as auto-generated transcriptions. Our experimental design includes two main benchmarking evaluations and a set of targeted ablation

¹Data are publicly available at this Github Repository.

²This guided strategy was adopted because preliminary tests showed that the model occasionally generated interpretive sentences instead of empty or low-confidence outputs

Model	Role
Qwen2.5-Omni-7B [5]	Main Experiment with Audio; Ablation Study
Qwen2.5-VL-7B [26]	Main Experiment with Transcriptions
Whisper-1 [24]	Audio transcription (Data enrichment)
GPT-4o-transcribe [25]	Audio transcription (Data enrichment)

Table 2

Models used in the experimental evaluation and audio-transcription pipeline.

studies. While the main experiments measure VLM performance in the intended multimodal setting, the ablations systematically modify the available input information to investigate the contribution of different informational channels.

4.1. Models

We used two VLMs for our experiments: Qwen2.5-VL-7B and Qwen2.5-Omni-7B. Qwen2.5-VL-7B is used both to replicate the original MAIA evaluation setting for the VSV task, and in the transcription-based experiments that extend this setup with automatically generated textual information. Qwen2.5-Omni-7B is used in the main audio-vision experiment, as well as in a series of ablation studies. Across all experiments, the visual input was processed consistently: 32 frames were uniformly sampled from each video and provided to the model as a sequence of independent static images. Table 2 summarizes their roles and also reports the speech-to-text models used in the audio-transcription pipeline described in Section 3. Further architectural details on the VLMs are provided in Appendix B.

4.2. Benchmarking VLMs with Audio

To assess how audio integration affects multimodal reasoning, we evaluated Qwen2.5-Omni-7B using the setup in Section 4.1, where each video is represented by 32 uniformly sampled frames. We also classify the raw 16 kHz audio into four categories: Dialogue, Unknown Dialogue, Music, and Silence/Noise, without preprocessing, normalization, or silence removal. This allows us to analyze performance across different acoustic contexts. Since visual input is provided as discrete frames rather than a continuous video tensor, no temporal alignment is applied. Audio, frames, and text prompts are passed directly to the multimodal processor and fused into a unified token sequence.³

To isolate the contribution of the audio modality, each video was processed twice: once with both visual frames and raw audio waveform provided to the model, and once with the audio component disabled, passing only the visual frames. This paired evaluation allows us to directly quantify the effect of audio integration on the model’s reasoning accuracy.

The primary goal of this experiment is to evaluate how audio integration affects model reasoning performance in the original VSV task in Testa et al. [10], under different acoustic conditions. The model must select the true statement from a true–false pair based on a specific visual and auditory context. The task is formulated as a strict multiple-choice prompt with the following structure: *Choose the correct description: A. [Statement] B. [Statement]. Answer only A or B.*, where the correct and incorrect options are randomly assigned to placeholders A and B. For example, for a video showing someone preparing pizza in a pizzeria, the two alternatives could be: A. *The mozzarella is melted by the heat of the wood-fired oven.* and B. *The mozzarella is melted by the heat generated by the sun.* Performance is evaluated based on standard accuracy, calculating the overall percentage of multiple-choice questions the model answers correctly across the dataset.

Ablation Study. To complement the main benchmarking experiment, we conduct a set of ablation studies aimed at providing a more diagnostic view of model behavior. These analyses examine how performance changes when the informational content available to the model is systematically controlled.

³All experiments use zero-shot inference with the Hugging Face Transformers library.

Experiment	Video	Transcription	Role
Unimodal Baseline	No	No	Bias baseline , question without any context
Transcription-only	No	Yes	Question with transcript-only context
32 Frames (without Transcription)	Yes	No	Question with video-only context
32 Frames (with Transcription)	Yes	Yes	Question with video plus transcription context

Table 3

Experimental conditions designed to explore how textual audio transcriptions prompts influence the reasoning capabilities of the Qwen2.5-VL-7B model.

This is particularly relevant in caption-foil settings, where a correct prediction may reflect different factors, including visual grounding, linguistic plausibility, or auxiliary textual evidence. Thus, ablation studies serve to disentangle the contribution of different input channels and to interpret the main results beyond aggregate performance scores.

To assess model reasoning under restricted visual contexts, we implemented two distinct frame settings, each tested systematically with audio (integrating 16 kHz raw waveforms) and without audio (visual-only baseline).

- **Black Frame Setting.** All visual input is replaced by a single 224x224 fully black image. This serves as a baseline to assess how well the model performs when relying entirely on audio and text, isolating the utility of audio when visual cues are completely absent.
- **1-Frame Setting.** Only the first frame of each video is extracted and provided to the model. This configuration removes temporal visual motion, testing whether a single static visual glimpse combined with temporal audio is sufficient for accurate reasoning.

For both setups, Qwen2.5-Omni-7B performs the same VSV task using the standardized prompt presented in Section 4.2. The visual input and raw waveform (when audio is enabled) are jointly provided to the model, allowing it to generate predictions from the available visual, auditory, and textual information.

4.3. Benchmarking VLMs with Transcriptions

While the previous section focused on raw audio, we further examine how speech semantics affect model performance. We evaluate Qwen2.5-VL-7B, a VLM without native audio-processing capabilities, by providing audio information indirectly through auto-generated transcriptions (Section 3.1). This setup separates the acoustic signal from its linguistic content, allowing us to test whether speech-derived semantics support video-based reasoning without the original waveform. Using this textual audio layer, we design controlled experiments to disentangle the roles of visual evidence, transcription-based information, and answer-option priors. This directly addresses *modality biases* in video-question answering, where models may rely on linguistic patterns or dataset regularities rather than visual grounding [27].

Our experimental conditions were designed to evaluate performance by isolating different information sources, as reported in Table 3. The experimental design consists of four main input configurations:

- **Unimodal Baseline.** The model receives only the VSV prompt formulation (i.e., *Choose the correct description*) and the two candidate captions, with no video or audio-derived text. This unimodal baseline assesses how much of the task can be addressed using the linguistic structure of the input alone, revealing potential biases in the answer options and their linguistic plausibility.
- **Transcription-only.** The model receives the transcription but no visual input. This condition isolates the contribution of textual speech information and evaluates whether it can support the decision-making process independently.
- **32 Frames (without Transcription).** The model receives only the visual input. This visual baseline assesses whether the task can be solved through visual grounding by aligning the candidate captions with the video content.

- **32 Frames (with Transcription).** The model receives both the video and the corresponding transcription. This condition assesses whether the transcription provides useful supplementary information when visual input is already available.

Together, these four conditions establish a structured framework designed to disentangle and analyze three distinct sources of information: patterns within answer options, insights from the transcription, and visual grounding evidence.

5. Results and Discussion

As discussed in Section 1, the MAIA dataset was designed around visually grounded reasoning, with most task-relevant information conveyed through the visual channel. However, previous results [10] highlighted persistent challenges in reasoning contexts such as temporal and implicit understanding, raising the question of whether audio may provide complementary evidence when visual information alone is not sufficient. Table 4 reports the results for both main experiments: the audio integration experiment (Qwen2.5-Omni-7B) and the transcription-based experiment (Qwen2.5-VL-7B), with a particular split per audio category, together with the respective Δ values measuring the degree of improvement or degradation introduced by each additional modality.

To determine whether these accuracy differences reflect systematic changes in item-level predictions, we additionally conducted two-sided exact McNemar tests on paired correctness outcomes [28], applying Holm correction to the comparisons across acoustic categories. Full results, including discordant prediction transitions, adjusted p -values, and confidence intervals, are reported in Appendix C.

Visual Dominance. The primacy of the visual stream over the audio emerges across both models, although the statistical patterns differ between raw audio and audio-derived transcriptions. For Qwen2.5-Omni-7B, the overall difference between the 32-frame vision-only and vision+audio configurations is negligible (-0.02 percentage points) and not statistically significant ($p = .620$; Appendix C, Table 9). This indicates that raw audio has no systematic effect on performance when dense visual evidence is available. For Qwen2.5-VL-7B, the 32 Frames (without Transcription) condition achieves 80.67% accuracy, compared with 78.78% when transcription is added. The resulting decrease of 1.89 percentage points is highly significant according to the item-wise McNemar test ($p = 3.60 \times 10^{-17}$), and remains significant within every acoustic category after Holm correction (Appendix C, Table 10). The video-only condition also represents the strongest configuration for this model, confirming the visual stream as its primary and most reliable source of task-relevant information, consistent with MAIA’s emphasis on fine-grained visual grounding [9, 10]. Taken together, these results show that visual evidence alone is sufficient to drive strong performance on this task. When dense visual information is available, raw audio produces no systematic change, whereas audio-derived transcriptions lead to a modest but statistically robust performance decrease.

Linguistic Bias and Unimodal Shortcuts. Before examining the effect of combining modalities, the Unimodal Baseline and Transcription-only conditions in the VL transcription-based experiments reveal an important dynamic. The Unimodal Baseline achieves 68.72% accuracy, demonstrating that the model can address a non-trivial portion of the VSV task through linguistic plausibility alone, without any visual or audio-derived input. More revealing is the Transcription-only condition, which at 63.98% performs below the Unimodal Baseline. This counterintuitive result indicates that transcriptions, rather than providing useful semantic grounding, introduce a weak and often misleading textual signal that interferes with the model’s reasoning. Speech-derived text is poorly aligned with the visual distinctions the task requires, and this holds even for videos labelled as `dialogue`, where understandable speech is present and the transcription could reasonably be expected to provide at least partial task-relevant content. This suggests that Qwen2.5-VL-7B is prone to assigning disproportionate weight to the textual channel even when its informative value is low or negative, a tendency that foreshadows the modality bias observed in the multimodal conditions.

Audio and Transcriptions under Dense Visual Evidence. When the additional modality is combined with 32 visual frames, raw audio and audio-derived transcriptions produce different statistical patterns.

For Qwen2.5-VL-7B, adding transcription reduces accuracy from 80.67% to 78.78%, a drop of 1.89 percentage points. The item-wise McNemar test confirms that this decrease is systematic ($p = 3.60 \times 10^{-17}$), and the effect remains significant in every acoustic category after Holm correction (Appendix C, Table 10). The effect is also confirmed at the video level through paired permutation tests (Appendix C). The audio-label breakdown further reinforces this result: the reduction ranges from `dialogue` (−1.16) and `unknown_dialogue` (−1.91) to `silence/noise` (−2.26) and `music` (−2.63). Thus, although moderate in magnitude, the detrimental effect of transcription is systematic rather than being driven by a small number of isolated prediction changes. By contrast, adding raw audio to Qwen2.5-Omni-7B produces no significant overall change (−0.02 points, $p = .620$), and none of the differences across acoustic categories remains significant after Holm correction. Therefore, under dense visual evidence, audio-derived transcriptions produce a modest but systematic interference effect, whereas raw audio is largely neutral rather than clearly harmful.

The Dialogue Paradox. The most counterintuitive finding concerns the `dialogue` category, which represents the condition most likely to benefit from spoken information. For Qwen2.5-Omni-7B, adding raw audio produces a small negative difference (−0.29 points); however, this change is not statistically significant after Holm correction ($p_{\text{Holm}} = 1.00$). It should therefore not be interpreted as evidence of systematic cross-modal interference. A different conclusion applies to Qwen2.5-VL-7B. Although `Dialogue` shows the smallest transcription-related decrease (−1.16 points), the effect remains statistically significant after correction ($p_{\text{Holm}} = .005$). Thus, even clear transcribed speech does not systematically improve VSV performance when dense visual evidence is already available, possibly because its content is not always informative for the specific caption–foil distinction. The only positive numerical trend is observed in the Omni setting for `music` (+1.07), which may suggest that background musical cues occasionally provide contextual information even when rich visual evidence is available. However, this difference does not remain statistically significant ($p_{\text{Holm}} = .225$). It should therefore be interpreted as a descriptive tendency rather than as reliable evidence of a beneficial effect. Taken together, the results provide evidence of a systematic modality interference effect in the transcription-based setting, consistently with modality bias [26, 27]. By contrast, the heterogeneous but non-significant category-level differences observed for Qwen2.5-Omni-7B suggest that raw audio may interact differently with distinct acoustic contexts. In particular, the positive trend observed for `music` may motivate further investigation into whether non-verbal acoustic cues are integrated differently from speech or noise, although the present results do not support any systematic effect.

Overall, a model with effective multimodal integration should preserve video-only performance when an uninformative signal is added. The statistical analyses show that Qwen2.5-VL-7B does not fully satisfy this expectation: adding transcription produces a moderate but highly systematic degradation, suggesting that audio-derived text can interfere with visually grounded decisions. For Qwen2.5-Omni-7B, however, the near-zero and non-significant 32-frame difference indicates that raw audio is largely ignored or suppressed when rich visual evidence is available.

The contrast between the significant −1.89-point transcription effect and the non-significant −0.02-point raw-audio effect therefore suggests that textualized audio is more likely than the raw acoustic stream to interfere with model predictions when its content is structurally misaligned with the caption–foil distinction.

5.1. Qualitative Analysis

To clarify the aggregate trends, we conducted a qualitative case-study analysis on the most extreme outcomes in both settings: raw-audio integration with Qwen2.5-Omni-7B and transcription-based augmentation with Qwen2.5-VL-7B. Videos were ranked by the performance change induced by the acoustic signal, selecting the largest improvement and the largest drop in each setting. This yielded two shared case studies, both from the `dialogue` category, where intelligible speech offers a richer acoustic signal. For each case, we report VSV items whose prediction changes are directly linked to audio-derived information.

Largest improvement. We selected *Video37* as it shows the largest performance gain when the acoustic

Model	Setting	Dialogue	Unknown Dialogue	Music	Silence/Noise	Avg.
Qwen2.5-Omni-7B	32 Frames (without Audio)	83.09%	81.73%	82.29%	83.70%	82.70%
	32 Frames (with Audio)	82.80%	81.29%	83.36%	83.28%	82.68%
	Δ (Audio – No Audio)	-0.29%	-0.44%	+1.07%	-0.42%	-0.02%
Qwen2.5-VL-7B	Unimodal Baseline	68.48%	67.84%	69.58%	68.84%	68.72%
	Transcription-only	66.95%	62.11%	62.62%	62.37%	63.98%
	32 Frames (without Transcription)	80.74%	78.99%	81.31%	80.86%	80.67%
	32 Frames (with Transcription)	79.58%	77.08%	78.68%	78.60%	78.78%
	Δ (Transcription - No Transcription)	-1.16	-1.91	-2.63	-2.26	-1.89

Table 4

Accuracy (%) grouped by audio category for two models. *Top section (Qwen2.5-Omni-7B)*: results for the 32-frame configuration with and without audio input; Δ reports the difference between audio-enabled and vision-only settings. *Bottom section (Qwen2.5-VL-7B)*: VSV accuracy across four experimental conditions; Δ reports the difference between the 32 Frames (with Transcription) and 32 Frames (without Transcription) settings. Positive Δ values indicate improvement, negative values indicate degradation.

signal is added, with raw audio yielding a delta of +2.60 and transcription a more modest +1.56. These gains correspond respectively to 18 and 8 Wrong→Correct transitions, i.e., cases in which the addition of the acoustic signal led the model to select the correct caption rather than the foil (Table 5). The following examples illustrate such cases in which the acoustic signal supports the model’s prediction.⁴

32 Frames (with Audio):

Question: *When does the little girl start crying?*

Caption: *The little girl starts crying when the other one makes her fall.*

Foil: *The little girl starts crying after the other one pinches her.*

32 Frames (with Transcription)

Question: *Why does the girl in the pink hoodie at the beginning of the video try to hold on to the wall?*

Caption: *The little girl tries to hold on to the wall to avoid falling.*

Foil: *The little girl tries to hold on to the wall to improve her balance.*

Transcription excerpt: *"you got hurt", "Vanessa, what a fall you took!", "she cut her hand", "The leg".*

This video shows a child falling and getting injured, while others react. Both raw audio and transcription align with the video’s core narrative: the fall, the injury, and the emotional response. Raw audio helps link the onset of crying to the fall, rather than to a later unrelated action. The transcription makes this interpretation explicit through phrases such as *you got hurt*, *what a fall you took*, and *she cut her hand*. Thus, audio does not replace visual grounding, but provides an auxiliary cue when the VSV contrast depends on causal, temporal, or affective interpretation.

Video	Δ_{audio}	Δ_{trans}	Wrong→Correct (audio)	Wrong→Correct (trans)
Video37	+2.60	+1.56	18	8

Table 5

Case study analysis for Video37 (larg. improvement). Δ_{audio} and Δ_{trans} report the overall accuracy change in % points; Wrong→Correct counts items that shifted from incorrect to correct after adding the acoustic signal.

Largest drop. The largest performance degradation is observed instead for Video71, with transcription yielding a delta of −5.73 and raw audio a slightly smaller drop of −4.17. These decreases correspond respectively to 15 and 12 Correct→Wrong transitions.

The following examples illustrate cases in which the acoustic signal reduces the model’s prediction:

32 Frames (with Audio):

Question: *What happens while the man is weighing the fruit?*

⁴Examples are originally in Italian. We provide English translations here.

Caption: *The man talks with the man wearing the flat cap.*

Foil: *After the man has weighed the fruit, he talks with the man wearing the flat cap.*

32 Frames (with Transcription):

Question: *What tool does the grey-haired man use in the video?*

Caption: *The grey-haired man in the video uses a scale.*

Foil: *The grey-haired man in the video uses a telescope.*

Transcription excerpt: *“let’s go with the oranges”, “they are good”, “they are beautiful and sweet”.*

This video depicts a street vendor selling oranges. Unlike Video37, the acoustic signal is not unrelated to the scene: it is coherent with the general commercial context, repeatedly referring to oranges and their qualities. However, this coherence is too generic to support the fine-grained distinctions required by the VSV items. In the raw-audio example, the key issue is temporal alignment: the audio confirms speech but not whether it occurs *while* the man weighs the fruit or *after*. In the transcription example, the transcript mentions oranges but not the tool, providing a plausible yet non-discriminative cue that weakens visual grounding. Thus, Video71 shows stronger modality interference than Video37: scene-coherent but item-insufficient audio leads the model to over-weight non-discriminative signals and lose visual grounding across temporal, spatial, tool-use, and object-localization items.

Video	Δ_{audio}	Δ_{trans}	Correct→Wrong (audio)	Correct→Wrong (trans)
Video71	-4.17	-5.73	12	15

Table 6

Case study analysis for Video71 (largest drop). Δ_{audio} and Δ_{trans} report the overall accuracy change in percentage points; Correct→Wrong counts items that shifted from correct to incorrect after adding the acoustic signal.

Overall, the comparison between Video37 and Video71 shows that audio is beneficial only when its content is directly aligned with the discriminative information required by the task, and harmful when it is only globally coherent with the scene without encoding the fine-grained distinction the VSV contrast demands. This pattern holds across both raw audio and transcription, although the degradation is stronger in the transcription-based setting, suggesting that textualized audio is particularly prone to attracting model attention even when it carries no task-relevant information.

5.2. Ablation Study: The Black Frame Phenomenon and the Visual Threshold

Table 7 reports the results of the ablation study (Qwen2.5-Omni-7B) under two visual configurations: Black Frame, where the visual input is replaced by a non-informative black frame, and 1-Frame, where a single static image is supplied.

The utility of audio is most pronounced in the Black Frame setting, where the absence of informative visual input forces the model to rely on the acoustic signal. Here, audio integration drives a substantial aggregate improvement of +5.79% (67.40% → 73.19%), confirming that raw audio carries genuine and independent semantic information when visual evidence is unavailable. The improvement is strongly supported by the paired analysis: the overall McNemar test is highly significant ($p = 2.44 \times 10^{-76}$), and all four category-level gains remain significant after Holm correction (Appendix C, Table 12). The gain is largest for *dialogue* (+7.68 points), followed by *silence/noise* (+6.12), *unknown_dialogue* (+5.77), and *music* (+3.62). The relatively smaller gain for *music* is coherent with the nature of the category: background music carries less task-relevant semantic content compared to intelligible speech, even in the absence of visual grounding. Transitioning to the 1-Frame setting reveals a sharp reduction in the measurable contribution of audio. The introduction of just a single visual frame is sufficient to make the audio signal largely redundant. The aggregate difference decreases to only +0.56 percentage points (78.84% → 79.40%). Although the aggregate item-wise McNemar test reaches nominal significance ($p = .0315$), the corresponding confidence interval includes zero and no acoustic category remains

significant after Holm correction (Appendix C, Table 11). In particular, the +0.45-point difference observed for dialogue is not significant ($p_{\text{Holm}} = .714$).

Therefore, the Black-Frame results provide robust evidence that audio supports performance when visual evidence is absent, whereas the 1-Frame results provide no robust category-level evidence of an additional audio benefit. This contrast supports the visual-threshold interpretation: once even minimal visual grounding is available, the marginal contribution of audio becomes substantially weaker.

Model	Setting	Dialogue	Unknown Dialogue	Music	Silence/Noise	Avg.
Black Frame	Black Frame (without Audio)	65.82%	65.80%	69.94%	68.03%	67.40%
	Black Frame (with Audio)	73.50%	71.57%	73.56%	74.15%	73.19%
	Δ (Audio – No Audio)	+7.68%	+5.77%	+3.62%	+6.12%	+5.79%
1-Frame	1-Frame (without Audio)	77.83%	77.60%	79.53%	80.40%	78.84%
	1-Frame (with Audio)	78.28%	77.69%	80.64%	80.98%	79.40%
	Δ (Audio – No Audio)	+0.45%	+0.09%	+1.11%	+0.58%	+0.56%

Table 7

Accuracy (%) grouped by audio category for two configurations of Qwen2.5-Omni-7B. *Top section (Black Frame)*: results for the black-frame configuration with and without audio input. *Bottom section (1-Frame)*: results for the 1-frame configuration with and without audio input. The Δ rows report the performance difference between the audio-enabled and vision-only settings; positive values indicate that audio improves performance.

5.3. Attention Allocation and the Limits of Implicit Fusion

To further investigate the mechanisms underlying the performance trends observed across configurations, we conducted an attention analysis on Qwen2.5-Omni-7B, examining how the model distributes its computational budget across modalities as visual density varies. To extract attention weights, we registered a forward hook on the final self-attention layer of Qwen2.5-Omni-7B’s thinker module (with `attn_implementation="eager"` to materialize attention tensors) and averaged the last generated token’s attention across all heads, partitioning the sequence into video, audio, and text regions via the model’s special boundary tokens (`<|vision_bos|>`, `<|audio_bos|>`, etc.).

We conducted this attention analysis across Black Frame, 1-, 12-, and 16-Frame configurations.⁵ Text consistently receives most of the attention budget (80-96%), but adding audio reallocates 11-14% of attention away from text. Audio attention is highest with Black Frames (13-14%), decreases with 1 Frame (11-12%), and progressively declines as visual density increases. This pattern suggests that the model implicitly adjusts the relevance of audio based on the amount of available visual information. However, greater audio attention does not necessarily improve reasoning. In the 1-Frame setting, audio receives 11 – 12% of the attention budget, despite producing only a small and statistically non-robust performance difference. This indicates that allocating attention to acoustic information does not necessarily translate into improved reasoning.

Overall, audio is most useful when visual evidence is absent and becomes increasingly marginal as visual density grows. Together with the transcription experiments on Qwen2.5-VL-7B, these findings point to a limitation of implicit multimodal fusion: current architectures do not effectively gate modality contributions according to their contextual informativeness.

6. Conclusion

In this paper, we presented a preliminary investigation into the role of audio information in video-based Visual Statement Verification task, evaluating two models from the Qwen2.5 family through a series of controlled experiments designed to isolate the contribution of different input modalities. Audio

⁵The 32-Frame setting was excluded due to out-of-memory constraints

information was examined in two complementary forms: as a raw acoustic signal directly processed by an omnimodal model, and as auto-generated transcriptions provided as auxiliary textual input to a vision-language model. Our results converge on three main findings: (i) visual evidence remains the primary source of task-relevant information, even when potentially informative auditory signals are explicitly made available; (ii) under dense visual evidence, raw audio produces no systematic performance change, whereas transcriptions induce a moderate but statistically significant degradation, pointing to a modality interference issue particularly evident in the transcription-based setting; (iii) raw-audio utility decreases sharply as visual information becomes available, providing a substantial and statistically robust benefit when visual evidence is absent, but only a small and non-robust contribution with one frame and a negligible effect with dense visual input. This pattern is also consistent with the attention analysis, which shows that the model implicitly modulates audio relevance according to visual availability, although attention to audio does not necessarily translate into improved reasoning. Whether these results reflect the visually grounded nature of the dataset or a more general difficulty of current multimodal architectures in balancing modalities remains an open question. Future work should replicate this analysis on a broader range of models and on datasets where audio carries genuinely task-critical information, in order to disentangle dataset-level from architecture-level contributions to the observed behavior.

7. Limitations

Our study is limited to two relatively small models belonging to the same model family. The findings should therefore not be interpreted as a general characterization of audio integration across omnimodal models. At the same time, the number of publicly available models that can process audio alongside visual and textual inputs remains relatively limited. Furthermore, evaluating models from the same family allowed us to examine how performance changes when a baseline model evolves to incorporate audio capabilities while being tested on the same data. Moreover, MAIA was not originally designed to evaluate audio-visual reasoning: while its videos provide a suitable test bed for an initial investigation, the relevance of the audio channel varies across instances. Future work should extend the analysis to a broader range of models and replicate the experimental protocol on additional video-based datasets specifically suited for studying the contribution of audio information.

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Declaration on Generative AI

During the preparation of this work, the authors used ChatGPT in order to: Grammar and spelling check. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication’s content.

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A. Data enrichment: Audio Transcription

Classification	Number of videos	Percentage
Dialogue	35	35%
Unknown_dialogue	12	12%
Music	17	17%
Silence/Noise	36	36%

Table 8

Distribution of the final audio-transcription classes assigned to the 100 videos after manual revision.

To construct this auxiliary textual channel, in order to enrich each video instance with a structured textual representation of its audio track, we constructed an auxiliary textual channel capturing both the transcribed content and a coarse characterization of the underlying acoustic signal. To this end, we automatically transcribed the audio tracks of the videos and assigned each instance a contextual label reflecting the nature of the audio signal. This process was then integrated into our experimental pipeline (see Section 3.1). To build this textual representation, we processed the audio track of each video with two automatic systems, *Whisper-1* [24] and *GPT-4o-transcribe* [25], which were used in complementary ways.

Whisper-1. This first system, was employed as a standard automatic speech recognition model, without any task-specific prompt. Whisper-1 is a speech-to-text model that supports transcription output in verbose JSON format, a response structure that includes not only the full transcript, but also segment-level metadata and, optionally, word-level timing information. In our setup, this format was particularly useful because it exposed a sequence of transcription segments, each associated with diagnostic fields such as log probability for each token, compression ratio, and no-speech probability. These metrics can be used to assess transcription quality and to detect potentially problematic segments, for example, when the average log probability is very low, the compression ratio is unusually high, or the probability of no speech is elevated. For each video, we therefore aggregated these segment-level values by averaging them across all segments, thereby deriving three global indicators: (i) a mean transcription log-probability score, (ii) a mean compression ratio, and (iii) a mean no-speech probability. These quantities were then used both as confidence indicators and as diagnostic signals for the contextual interpretation of the audio track. On the basis of these aggregated values and of simple lexical heuristics applied to the transcription, we defined the following four contextual classes, which were then assigned to the Whisper-1 output:

- **MUSIC.** Instances characterized by repetitive vocalizations, onomatopoeic patterns, strongly repetitive tokens, or explicit lexical cues associated with non-speech audio.
- **SILENCE/NOISE.** Videos with empty or highly unreliable transcriptions, especially when accompanied by low log-probability values and relatively high no-speech probability.
- **DIALOGUE.** Videos whose transcriptions displayed sufficiently coherent linguistic content, acceptable log-probability values, low degeneracy, and limited evidence of no-speech segments.
- **UNKNOWN_DIALOGUE.** Residual cases that still appeared speech-related but did not satisfy the dialogue criteria with sufficient confidence were assigned to this category, thus preserving potentially informative content while explicitly marking it as uncertain.

GPT-4o-transcribe. The second branch of this transcription pipeline followed a different logic. GPT-4o-transcribe is a speech-to-text model that returns transcription output in JSON format and can additionally expose token-level log probabilities when these are explicitly requested through the corresponding include parameter. This makes it particularly suitable not only for generating a transcript, but also for associating the generated output with a direct confidence signal derived from the model’s own token-level probabilities. In our pipeline, however, the model was not used as a purely neutral transcriber. Instead, it was employed as a guided transcription-and-classification system. More specifically, it was explicitly instructed to output the label `MUSIC` when the audio contained music, `SILENCE/NOISE` when the track contained only noise or other non-informative acoustic content, and `UNKNOWN_DIALOGUE` when the audio contained dialogue that could not be reliably understood. Only in the presence of sufficiently intelligible spoken content the model was expected to produce an actual transcription⁶. In addition, token-level log probabilities were extracted from the output and averaged in order to obtain a global confidence estimate for the entire prediction. The GPT-based outputs were then normalized into the same four contextual classes used for Whisper-1. Whenever the returned text was one of the coarse labels, the transcription field was left empty, otherwise, the returned text was treated as a transcription and the contextual label was mapped to dialogue.

Merging Whisper and GPT4o transcriptions. The outputs of the two systems were subsequently merged through a dedicated decision procedure. When Whisper-1 and GPT-4o-transcribe agreed on the same contextual class, that class was retained as the final one. In agreement cases, the reliability of the final annotation was estimated by combining the confidence signals produced by the two models. Whisper’s contribution was not defined through a single fixed formulation from the outset. Instead, several alternative scoring and weighting configurations were tested in order to arrive at a more stable and interpretable reliability measure. The final formulation adjusted Whisper’s average log-probability through a reliability score that incorporated two penalty terms: one proportional to the no-speech probability, and another activated only when the compression ratio exceeded a conservative threshold. This made it possible to downgrade outputs that were likely to be unreliable, silent, or affected by degenerate transcription behavior. The resulting Whisper score was then combined with the GPT-4o-transcribe score through a weighted average, assigning 45% of the final contribution to Whisper and 55% to GPT-4o-transcribe. This weighting scheme was selected after iterative experimentation and reflects a compromise between the diagnostic value of Whisper’s acoustic indicators and the stronger linguistically guided interpretation provided by GPT-4o-transcribe. When the agreed contextual class was `DIALOGUE` or `UNKNOWN_DIALOGUE`, the pipeline also retained a transcription. In these cases, the textual outputs of the two systems were not simply concatenated. Instead, they were merged through a semantic reconciliation step. First, the two full transcriptions were compared globally: if they were judged to be near-duplicates, only the better textual variant was kept. Otherwise, they were segmented into sentences and merged incrementally, with semantically redundant material removed and better variants preserved whenever two sentences expressed substantially the same content. This allowed us to exploit the complementarity of the two transcription systems while minimizing noise and duplication in the final text that would later be used for the experiments. When disagreements emerged between the two systems regarding contextual classification, a distinct strategy was employed. In such cases, the final label was determined based on the highest reliability score. GPT-4o-transcribe’s reliability was measured using its average token-level log probability, while Whisper-1 benefited from additional indicators, including its no-speech probability and compression metrics. After selecting a model, its output reliability was further analyzed using a four-tier threshold system: outputs with scores of -0.3

⁶This design choice was motivated by an empirical observation made during the preliminary inspection of the outputs. In several videos whose audio contained only music, the model did not behave as a simple low-level transcriber, but generated interpretable sentences such as “*La musica è bella*”. This suggested that GPT-4o-transcribe was not merely converting sound into text at a surface level, but was already capable of forming an internal interpretation of whether the input corresponded to actual dialogue or to some other kind of audio content. On this basis, we treated it directly as a first-pass classifier, rather than relying exclusively on downstream heuristic classification based on confidence metrics as done before.

or higher were classified as highly reliable, scores between -0.6 and -0.3 as reliable, values from -0.9 to -0.6 as uncertain, and scores below -0.9 as unreliable. Beyond these scoring categories, two specific override conditions were established for Whisper to enhance accuracy. Firstly, when the no-speech probability exceeded 0.8, the output was automatically flagged as unreliable, regardless of its calculated score. Secondly, if the compression ratio surpassed 2.4, the transcription was marked as unreliable due to indications of instability or flawed generation. To ensure the precision and consistency of the final dataset prepared for experiments, all outputs underwent manual review. This additional layer of verification focused on identifying residual errors, addressing problematic cases, and improving overall dataset quality for subsequent use. Table 8

B. Models

Qwen2.5-VL-7B We use *Qwen2.5-VL-7B*, a 7B vision-language model built on a Qwen2.5-based language model [29], a ViT visual encoder, and an MLP vision-language merger [26]. The model processes image and video inputs together with textual prompts, but does not support audio input. Its visual encoder adopts dynamic-resolution processing, representing images and video frames as variable-length visual token sequences, while its video processing module incorporates dynamic FPS sampling and time-aware multimodal positional encoding to support temporal modeling. The vision-language merger projects visual features into the language-model embedding space, enabling the decoder to integrate visual and textual information.

Qwen2.5-Omni-7B For the audio-vision experiments, we used *Qwen2.5-Omni-7B*, a 7-billion-parameter omnimodal model extending Qwen2.5-VL with a dedicated audio encoder and Thinker-Talker architecture [5]. The audio encoder processes raw waveforms into 128-channel mel-spectrograms at 40 ms resolution, while the ViT-based vision encoder handles static images and video sequences [26]. Its Time-aligned Multimodal RoPE (TMRoPE) enables explicit temporal alignment between audio and visual tokens: in video modality, frame i is grounded to audio at timestamp t_i , whereas image modality treats visual input as a static snapshot without temporal dependency [5]. We leverage only the Thinker’s text-generation capability for the VSV task, allowing direct comparison of audio integration effects across modality configurations

C. Statistical Significance of Audio and Transcription Effects

To assess whether the performance differences observed across input configurations reflect systematic changes in model predictions, rather than small fluctuations in aggregate accuracy, we conducted paired statistical tests at the item level. In particular, we applied two-sided exact McNemar tests to the binary correctness outcomes obtained for the same VSV items under each pair of experimental conditions.

For each comparison, we distinguish between items that changed from correct to incorrect after adding the additional modality ($C \rightarrow W$) and items that changed from incorrect to correct ($W \rightarrow C$). McNemar’s test evaluates whether these two types of discordant transitions occur with comparable frequency. Thus, unlike a comparison between two independent accuracy estimates, the test explicitly accounts for the paired nature of our experimental design.

We report the accuracy difference $\Delta = \text{Acc}_{\text{augmented}} - \text{Acc}_{\text{baseline}}$ in percentage points, the number of discordant transitions, exact two-sided p -values, and 95% confidence intervals for Δ . Since the analyses by acoustic context involve four simultaneous comparisons (Dialogue, Unknown Dialogue, Music, and Silence/Noise) we adjust category-level p -values using Holm’s procedure.

C.1. Dense Visual Evidence: 32-Frame Experiments

We first consider the experiments in which the model receives 32 uniformly sampled video frames. These comparisons assess whether raw audio or audio-derived transcriptions provide additional information

Table 9

Item-wise comparison of the Qwen2.5-Omni-7B 32-frame configurations without and with raw audio. Δ is expressed in percentage points. C $\rightarrow$ W denotes items answered correctly without audio and incorrectly with audio; W $\rightarrow$ C denotes the reverse transition. Category-level p -values are adjusted using Holm’s procedure.

Category	No audio	With audio	Δ	C $\rightarrow$ W	W $\rightarrow$ C	p	p_{Holm}	95% CI
Overall	82.70	82.68	-0.02	994	971	.61969	–	[−0.88, 0.64]
Dialogue	83.09	82.80	-0.29	379	360	.50788	1.00000	[−1.58, 1.00]
Unknown Dialogue	81.73	81.29	-0.44	120	110	.55289	1.00000	[−2.68, 1.81]
Music	82.29	83.36	+1.07	141	176	.05618	.22472	[−0.76, 2.90]
Silence/Noise	83.70	83.28	-0.42	354	325	.28258	.84774	[−1.66, 0.82]

when relatively dense visual evidence is already available.

Qwen2.5-Omni-7B: With vs. Without Raw Audio Table 9 reports the paired comparison between the 32-frame conditions with and without raw audio. At the aggregate level, adding audio changes accuracy from 82.70% to 82.68%, corresponding to a negligible decrease of 0.02 percentage points. The numbers of C $\rightarrow$ W and W $\rightarrow$ C transitions are similar (994 and 971, respectively), and the exact McNemar test provides no evidence of a systematic difference ($p = .620$). The confidence interval also includes zero ([−0.88, 0.64]).

No acoustic category shows a statistically significant change after Holm correction. Dialogue, Silence/Noise, and Unknown Dialogue exhibit small negative differences, whereas Music shows a positive difference of 1.07 points. However, even the Music effect does not reach conventional significance before correction ($p = .056$) and becomes clearly non-significant after correction ($p_{\text{Holm}} = .225$). Therefore, the raw-audio results do not support either a systematic improvement or a systematic degradation when rich visual evidence is available.

Qwen2.5-VL-7B: With vs. Without Transcriptions A different pattern emerges when audio information is converted into text and provided to Qwen2.5-VL-7B. As shown in Table 10, adding the transcription reduces overall accuracy from 80.67% to 78.78%, a difference of 1.89 percentage points. Among the discordant items, 1,116 predictions change from correct to incorrect, compared with 752 changes in the opposite direction. This asymmetry is highly significant according to the exact McNemar test ($p = 3.60 \times 10^{-17}$). Among items whose outcome changes, adding the transcription is therefore approximately 1.48 times more likely to damage a correct prediction than to repair an incorrect one.

The negative effect remains significant in every acoustic category after Holm correction. Dialogue presents the smallest reduction, at 1.16 points, but the effect is nevertheless significant ($p_{\text{Holm}} = .005$). Larger reductions are observed for Unknown Dialogue (−1.91), Silence/Noise (−2.26), and Music (−2.63). Furthermore, the confidence intervals are entirely below zero for the overall comparison and for each category.

These results indicate that the degradation produced by audio-derived transcriptions is not due only to a few isolated prediction changes. Rather, it constitutes a modest but systematic interference effect. Importantly, even the Dialogue category does not benefit from the additional textual content. Intelligible speech is therefore not necessarily useful unless its content directly encodes the fine-grained distinction between the caption and the foil.

Because several VSV items are associated with the same video, we also examined the transcription effect at the video level. Confidence intervals obtained by bootstrapping videos as clusters confirm the overall negative effect, while paired sign-flip permutation tests on per-video accuracy differences lead to the same conclusion. The overall permutation test yields $p < 10^{-6}$, and all four category-level comparisons remain significant after Holm correction. This additional analysis indicates that the transcription effect is not driven by a small number of individual videos.

Table 10

Item-wise comparison of the Qwen2.5-VL-7B 32-frame configurations without and with audio-derived transcriptions. Δ is expressed in percentage points.

Category	No trans.	With trans.	Δ	C→W	W→C	p	p_{Holm}	95% CI
Overall	80.67	78.78	-1.89	1,116	752	3.60×10^{-17}	–	[−2.39, −1.41]
Dialogue	80.74	79.58	-1.16	365	287	.00254	.00508	[−1.99, −0.31]
Unknown Dialogue	78.99	77.08	-1.91	125	81	.00265	.00508	[−3.08, −0.56]
Music	81.31	78.68	-2.63	205	119	2.05×10^{-6}	6.14×10^{-6}	[−3.80, −1.56]
Silence/Noise	80.86	78.60	-2.26	421	265	2.80×10^{-9}	1.12×10^{-8}	[−3.04, −1.46]

Table 11

Item-wise comparison of the Qwen2.5-Omni-7B 1-frame configurations without and with raw audio. Δ is expressed in percentage points.

Category	No audio	With audio	Δ	C→W	W→C	p	p_{Holm}	95% CI
Overall	78.84	79.40	+0.56	1,161	1,268	.03150	–	[−0.25, 1.38]
Dialogue	77.83	78.28	+0.45	448	477	.35724	.71448	[−0.98, 1.86]
Unknown Dialogue	77.60	77.69	+0.09	145	147	.95333	.95333	[−2.32, 2.49]
Music	79.53	80.64	+1.11	176	212	.07559	.30237	[−0.83, 3.04]
Silence/Noise	80.40	80.98	+0.58	392	432	.17426	.52279	[−0.74, 1.89]

C.2. Restricted Visual Evidence

We next examine whether the contribution of raw audio changes when the amount of visual information is reduced. We separately consider the 1-frame condition, which retains minimal visual grounding, and the Black-Frame condition, in which task-relevant visual evidence is absent.

One Frame: With vs. Without Raw Audio As reported in Table 11, adding raw audio in the 1-frame setting increases overall accuracy from 78.84% to 79.40%, a difference of only 0.56 percentage points. The item-wise McNemar test reaches nominal significance at the aggregate level ($p = .0315$), as 1, 268 items change from incorrect to correct and 1, 161 change from correct to incorrect. However, the 95% confidence interval includes zero ([−0.25, 1.38]), indicating that the magnitude and robustness of this small aggregate difference should not be overstated.

More importantly, no individual acoustic category shows a significant change after Holm correction. This directly addresses the Dialogue example highlighted in the review: accuracy increases by only 0.45 percentage points, from 77.83% to 78.28%, with 448 C→W and 477 W→C transitions. The corresponding McNemar test is not significant ($p = .357$; $p_{\text{Holm}} = .714$), and its confidence interval, [−0.98, 1.86], includes zero.

The same conclusion applies to Music, Silence/Noise, and Unknown Dialogue. Thus, while the aggregate item-wise comparison suggests a small positive tendency, the category-level results provide no robust evidence that audio systematically improves VSV performance once even a single visual frame is available.

Black Frame: With vs. Without Raw Audio The Black-Frame condition presents a substantially different pattern. In the absence of informative visual input, adding raw audio increases overall accuracy from 67.40% to 73.19%, corresponding to a gain of 5.79 percentage points. The number of W→C transitions is considerably larger than the number of C→W transitions (2, 603 vs. 1, 428), and the resulting McNemar test is highly significant ($p = 2.44 \times 10^{-76}$). Among discordant items, audio is approximately 1.82 times more likely to repair an incorrect prediction than to damage a correct one.

Unlike the 1-frame and 32-frame conditions, the improvement remains significant in all four acoustic categories after Holm correction. Dialogue shows the largest increase, at 7.68 percentage points, followed by Silence/Noise at 6.12 points, Unknown Dialogue at 5.77 points, and Music at 3.62 points.

Table 12

Item-wise comparison of the Qwen2.5-Omni-7B Black-Frame configurations without and with raw audio. Δ is expressed in percentage points.

Category	No audio	With audio	Δ	C→W	W→C	p	p_{Holm}	95% CI
Overall	67.40	73.19	+5.79	1,428	2,603	2.44×10^{-76}	–	[5.27, 7.10]
Dialogue	65.82	73.50	+7.68	497	998	2.99×10^{-38}	1.20×10^{-37}	[6.10, 9.25]
Unknown Dialogue	65.80	71.57	+5.77	167	300	1.01×10^{-9}	2.01×10^{-9}	[3.10, 8.45]
Music	69.94	73.56	+3.62	259	377	3.50×10^{-6}	3.50×10^{-6}	[1.43, 5.80]
Silence/Noise	68.03	74.15	+6.12	505	928	7.34×10^{-29}	2.20×10^{-28}	[4.61, 7.63]

All corresponding confidence intervals are above zero.

These results provide strong evidence that raw audio carries information that can support VSV predictions when visual evidence is unavailable. However, the positive results for coarse classes such as Silence/Noise should not be interpreted as showing that noise itself is necessarily task-relevant. These labels characterize the predominant acoustic context of a video and may still include contextual or low-level cues that the model can exploit.

C.3. Cross-Condition Interpretation

Taken together, the statistical analyses qualify the aggregate accuracy differences reported in the main paper. Raw audio has a substantial and statistically robust positive effect when visual evidence is absent, as in the Black-Frame condition. Its contribution becomes small and not robust across acoustic categories when only one frame is available, and no systematic effect is observed when 32 frames are provided. The transcription-based experiment exhibits a different behavior. Under the 32-frame configuration, audio-derived text produces a statistically significant decrease both overall and within every acoustic category. Therefore, while raw audio appears to be largely ignored when visual information is abundant, transcribed audio constitutes a more salient textual signal and can systematically interfere with visually grounded predictions.

Overall, these results support a visual-dominance interpretation, while also showing that statistical significance and effect magnitude must be considered separately. The Black-Frame improvement is both substantial and highly significant; the 1-frame difference is small and not robust at the category level; the 32-frame raw-audio effect is neither statistically significant nor practically meaningful; and the negative transcription effect is moderate in magnitude but highly systematic.